

# Business Analytics - Machine Learning Assignment: Bank Marketing Dataset Analysis

Github link: <https://github.com/dhanushhhh/ML-assignment>

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## Executive Summary

This assignment demonstrates the application of statistical and machine learning concepts learned in the Business Analytics course using the UCI Bank Marketing Dataset. The dataset contains 45,211 records from direct marketing campaigns of a Portuguese banking institution, with the objective of predicting whether a client will subscribe to a term deposit (binary target: yes/no).

The project integrates data classification, exploratory analysis, hypothesis testing, correlation analysis, sampling techniques, and supervised machine learning algorithms (logistic regression, decision trees, Naive Bayes, KNN) using R programming. Results are interpreted from three perspectives: statistician, data scientist, and marketing manager, with clear managerial implications for campaign optimisation and customer targeting.

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## 1. Introduction

### 1.1 Background

Direct marketing campaigns in banking are resource-intensive, requiring significant investment in call centre operations, agent time, and customer contact management. The Bank Marketing dataset provides an opportunity to apply predictive analytics to

improve campaign efficiency by identifying customers with higher subscription probability and reducing unproductive contacts.

## 1.2 Business Objective

From a managerial perspective, the core objectives are:

- Increase the conversion rate of term deposit subscriptions
- Reduce cost per acquisition by prioritising high-probability leads
- Optimise resource allocation across customer segments
- Understand key drivers of subscription behaviour

## 1.3 Analytical Objective

The analytical goal is to build and evaluate supervised classification models that predict subscription likelihood and translate model insights into actionable marketing strategies.

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# 2. Dataset Description

## 2.1 Data Source

**Dataset:** Bank Marketing Dataset  
**Repository:** UCI Machine Learning Repository (Dataset ID: 222)  
**File Used:** bank-full.csv  
**Records:** 45,211 instances  
**Variables:** 17 (16 input attributes + 1 binary target)  
**Task Type:** Supervised binary classification  
**Target Variable:** y (subscribed to term deposit: yes/no)

## 2.2 Variable Classification

The following table classifies all variables by data type, measurement scale, and role:

| Variable | Description | Qual/Quant | Measurement Scale | Continuous/Categorical | Variable Type |
|----------|-------------|------------|-------------------|------------------------|---------------|
|----------|-------------|------------|-------------------|------------------------|---------------|

|           |                                |              |         |             |             |
|-----------|--------------------------------|--------------|---------|-------------|-------------|
| age       | Client age (years)             | Quantitative | Ratio   | Continuous  | Independent |
| job       | Type of job                    | Qualitative  | Nominal | Categorical | Independent |
| marital   | Marital status                 | Qualitative  | Nominal | Categorical | Independent |
| education | Education level                | Qualitative  | Ordinal | Categorical | Independent |
| default   | Has credit in default?         | Qualitative  | Nominal | Categorical | Independent |
| balance   | Average yearly balance (euros) | Quantitative | Ratio   | Continuous  | Independent |
| housing   | Has housing loan?              | Qualitative  | Nominal | Categorical | Independent |
| loan      | Has personal loan?             | Qualitative  | Nominal | Categorical | Independent |
| contact   | Contact communication type     | Qualitative  | Nominal | Categorical | Independent |
| day       | Last contact day of month      | Quantitative | Ratio   | Discrete    | Independent |
| month     | Last contact month             | Qualitative  | Nominal | Categorical | Independent |

|          |                                    |              |         |                      |             |
|----------|------------------------------------|--------------|---------|----------------------|-------------|
| duration | Last contact duration (seconds)    | Quantitative | Ratio   | Continuous           | Independent |
| campaign | Number of contacts this campaign   | Quantitative | Ratio   | Discrete (count)     | Independent |
| pdays    | Days since last contact            | Quantitative | Ratio   | Discrete (count)     | Independent |
| previous | Number of contacts before campaign | Quantitative | Ratio   | Discrete (count)     | Independent |
| poutcome | Outcome of previous campaign       | Qualitative  | Nominal | Categorical          | Independent |
| y        | Subscribed to term deposit?        | Qualitative  | Nominal | Categorical (binary) | Dependent   |

## 2.3 Data Source Classification

From a data sources perspective:

**For the original bank:**

- Primary Internal Data: Customer records, campaign logs, transaction history
- Primary External Data: Direct customer responses during calls

**For this analysis (MBA student using UCI repository):**

- **Secondary External Data:** Dataset obtained from a public academic repository (UCI), originally collected by the Portuguese bank and shared for research purposes

**Additional context:**

- Secondary Internal (example): Historical campaign performance reports within the bank
  - Secondary External Government Data (example): National economic indicators, unemployment rates
  - Secondary External Syndicated Data (example): Industry benchmark reports from marketing research firms
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## **3. Data Understanding and Exploratory Analysis**

### **3.1 Data Loading and Initial Inspection**

**R Code:**

#### **Load libraries**

```
library(tidyverse)
library(caret)
library(e1071)
library(rpart)
library(rpart.plot)
library(class)
library(pROC)
```

#### **Load data**

```
bank <- read.csv("bank-full.csv", sep = ";")
str(bank)
summary(bank)
```

#### **Convert character variables to factors**

```
bank <- bank %>% mutate(across(where(is.character), as.factor))
```

```
> str(bank)
'data.frame': 45211 obs. of 17 variables:
 $ age      : int  58 44 33 47 33 35 28 42 58 43 ...
 $ job      : Factor w/ 12 levels "admin.", "blue-collar",...: 5 10 3 2 12 5 5 3 6 10 ...
 $ marital  : Factor w/ 3 levels "divorced", "married",...: 2 3 2 2 3 2 3 1 2 3 ...
 $ education: Factor w/ 4 levels "primary", "secondary",...: 3 2 2 4 4 3 3 3 1 2 ...
 $ default  : Factor w/ 2 levels "no", "yes": 1 1 1 1 1 1 1 2 1 1 ...
 $ balance  : int  2143 29 2 1506 1 231 447 2 121 593 ...
 $ housing  : Factor w/ 2 levels "no", "yes": 2 2 2 2 1 2 2 2 2 2 ...
 $ loan     : Factor w/ 2 levels "no", "yes": 1 1 2 1 1 1 2 1 1 1 ...
 $ contact  : Factor w/ 3 levels "cellular", "telephone",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ day      : int  5 5 5 5 5 5 5 5 5 5 ...
 $ month    : Factor w/ 12 levels "apr", "aug", "dec",...: 9 9 9 9 9 9 9 9 9 9 ...
 $ duration : int  261 151 76 92 198 139 217 380 50 55 ...
 $ campaign : int  1 1 1 1 1 1 1 1 1 1 ...
 $ pdays   : int  -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 ...
 $ previous : int  0 0 0 0 0 0 0 0 0 0 ...
 $ poutcome : Factor w/ 4 levels "failure", "other",...: 4 4 4 4 4 4 4 4 4 4 ...
 $ y        : Factor w/ 2 levels "no", "yes": 1 1 1 1 1 1 1 1 1 1 ...

> summary(bank)
   age      job      marital      education      default      balance
Min. :18.00  blue-collar:9732  divorced: 5207  primary : 6851  no :44396  Min. : -8019
1st Qu.:33.00  management :9458  married :27214  secondary:23202  yes: 815  1st Qu.: 72
Median :39.00  technician :7597  single :12790  tertiary :13301  Median : 448
Mean :40.94  admin. :5171  unknown : 1857  Mean : 1362
3rd Qu.:48.00  services :4154  Max. :102127
Max. :95.00  retired :2264
              (other) :6835

   housing  loan      contact      day      month      duration
no :20081  no :37967  cellular :29285  Min. : 1.00  may :13766  Min. : 0.0
yes:25130  yes: 7244  telephone: 2906  1st Qu.: 8.00  jul : 6895  1st Qu.: 103.0
              unknown :13020  Median :16.00  aug : 6247  Median : 180.0
              Mean :15.81  jun : 5341  Mean : 258.2
              3rd Qu.:21.00  nov : 3970  3rd Qu.: 319.0
              Max. :31.00  apr : 2932  Max. :4918.0
              (Other): 6060

   campaign      pdays      previous      poutcome      y
Min. : 1.000  Min. : -1.0  Min. : 0.0000  failure: 4901  no :39922
1st Qu.: 1.000  1st Qu.: -1.0  1st Qu.: 0.0000  other : 1840  yes: 5289
Median : 2.000  Median : -1.0  Median : 0.0000  success: 1511
Mean : 2.764  Mean : 40.2  Mean : 0.5803  unknown:36959
3rd Qu.: 3.000  3rd Qu.: -1.0  3rd Qu.: 0.0000
Max. :63.000  Max. :871.0  Max. :275.0000
```

## 3.2 Target Variable Distribution

R Code:

```
table(bank$y)prop.table(table(bank$y))
```

```
> table(bank$y)
   no   yes
39922 5289

> prop.table(table(bank$y))
   no      yes
0.8830152 0.1169848

> |
```

### Interpretation:

The dataset shows class imbalance with significantly more "no" responses than "yes" responses. This imbalance must be considered during model evaluation, making metrics like precision, recall, and F1-score more informative than raw accuracy.

## 3.3 Descriptive Statistics

### R Code:

```
numeric_vars <- c("age", "balance", "duration", "campaign", "pdays", "previous")
summary(bank[numeric_vars])
```

```
> numeric_vars <- c("age", "balance", "duration", "campaign", "pdays", "previous")
> summary(bank[numeric_vars])
      age      balance      duration      campaign      pdays      previous
Min.   :18.00   Min.   : -8019   Min.    :  0.0   Min.    :  1.000   Min.    : -1.0   Min.    :  0.0000
1st Qu.:33.00   1st Qu.:   72    1st Qu.: 103.0   1st Qu.:  1.000   1st Qu.: -1.0   1st Qu.:  0.0000
Median :39.00   Median :   448   Median : 180.0   Median :  2.000   Median : -1.0   Median :  0.0000
Mean   :40.94   Mean    : 1362   Mean    : 258.2   Mean    :  2.764   Mean    : 40.2   Mean    :  0.5803
3rd Qu.:48.00   3rd Qu.:  1428   3rd Qu.: 319.0   3rd Qu.:  3.000   3rd Qu.: -1.0   3rd Qu.:  0.0000
Max.   :95.00   Max.    :102127   Max.    :4918.0   Max.    :63.000   Max.    :871.0   Max.    :275.0000
> |
```

### Key Observations:

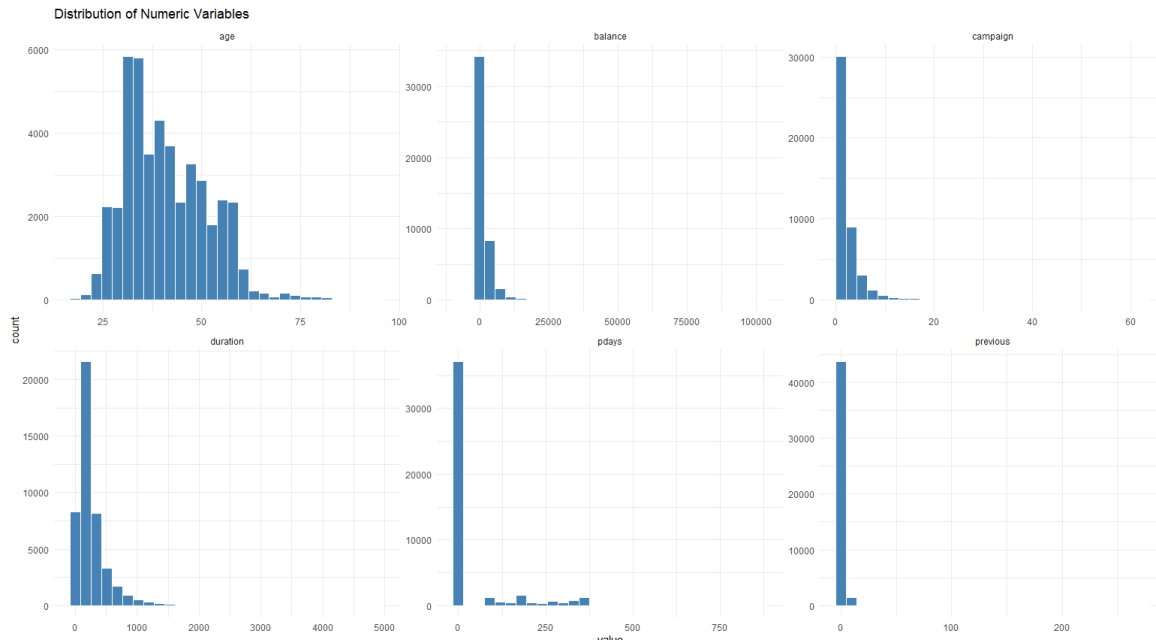
- Age ranges from **18 to 95** with mean **40.94**
- Balance is highly skewed with presence of outliers
- Duration varies widely, indicating different call engagement levels
- Campaign contacts typically range from **1 to 63**

## 3.4 Distributions and Normality

### R Code:

### Histograms for numeric variables

```
bank %>%
  select(all_of(numeric_vars)) %>%
  gather(variable, value) %>%
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~ variable, scales = "free") +
  theme_minimal() +
  labs(title = "Distribution of Numeric Variables")
```



### Distribution Analysis:

- Normal Distribution: **Age** approximately follows a normal distribution with a slight right skew
- Standard Normal Distribution: **Variables** can be standardized using z-scores for models requiring normalized inputs
- Poisson Distribution: **Campaign** and **previous** contacts show Poisson-like behavior as they are discrete count variables
- Skewed Distributions: **Balance** and **duration** show strong right skew, indicating the presence of extreme values

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## 4. Correlation Analysis

### 4.1 Pearson's r (Linear Correlation)

R Code:

**Pearson correlation matrix**

```
cor_pearson <- cor(bank[numeric_vars], use = "complete.obs", method = "pearson")
print(round(cor_pearson, 3))
```

```
> cor_pearson <- cor(bank[numeric_vars], method="pearson")
> round(cor_pearson, 3)
      age balance duration campaign  pdays previous
age      1.000  0.098  -0.005   0.005 -0.024   0.001
balance  0.098  1.000   0.022  -0.015  0.003   0.017
duration -0.005  0.022   1.000  -0.085 -0.002   0.001
campaign  0.005 -0.015  -0.085   1.000 -0.089  -0.033
pdays   -0.024  0.003  -0.002  -0.089  1.000   0.455
previous  0.001  0.017   0.001  -0.033  0.455   1.000
> |
```

### Interpretation:

Pearson's r measures linear relationships between continuous variables. Key findings:

- **Strongest positive correlations:** The strongest link is between **pdays** and **previous** ( $r = 0.455$ ), followed by a very weak link between **age** and **balance** ( $r = 0.098$ ).
- **Strongest negative correlations:** The most notable (though weak) negative relationship is between **campaign** and **duration** ( $r = -0.085$ ).
- **Note: correlation does not imply causation.** These values indicate the strength of association between features like age, balance, and contact frequency, not a direct cause.

## 4.2 Spearman's Rho (Rank Correlation)

R Code:

### Spearman correlation for ordinal relationships

```
cor_spearman <- cor(bank[numeric_vars], use = "complete.obs", method = "spearman")
print(round(cor_spearman, 3))
```

```
> cor_spearman <- cor(bank[numeric_vars], method="spearman")
> round(cor_spearman, 3)
      age balance duration campaign  pdays previous
age      1.000  0.096  -0.033   0.037 -0.017  -0.012
balance  0.096  1.000   0.043  -0.031  0.070   0.080
duration -0.033  0.043   1.000  -0.108  0.029   0.031
campaign  0.037 -0.031  -0.108   1.000 -0.112  -0.108
pdays   -0.017  0.070   0.029  -0.112  1.000   0.986
previous -0.012  0.080   0.031  -0.108  0.986   1.000
```

**Interpretation:**

Spearman's rho captures monotonic (not necessarily linear) relationships and is robust to outliers.

**Strongest positive:** **pdays** and **previous** (\$0.986\$) show a near-perfect rank-based relationship.

**Strongest negative:** **campaign** and **pdays** (\$-0.112\$) indicate a slight inverse ranking.

**Demographics:** **age** and **balance** (\$0.096\$) suggest older customers slightly favor higher balances.

**Note:** Correlation does not imply causation.

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## 5. Sampling Techniques and Central Limit Theorem

### 5.1 Train-Test Split with Stratified Sampling

**R Code:**

```
set.seed(123)
train_index <- createDataPartition(bank$y, p = 0.7, list = FALSE)
train <- bank[train_index, ]
test <- bank[-train_index, ]
```

### Verify stratification

```
prop.table(table(trainy))prop.table(table(testy))
```

```
> prop.table(table(train$y))
      no      yes
0.8829979 0.1170021
> |
```

**Sampling Concepts Applied:**

- **Random Sampling:** Each record has equal probability of selection
- **Stratified Random Sampling:** Sampling preserves the proportion of yes/no responses, critical given class imbalance

- **Central Limit Theorem:** With large sample size ( $n > 30$ ), sampling distribution of means approaches normal distribution, justifying use of z-tests and t-tests for inference
- 

## 6. Hypothesis Testing

### 6.1 Hypothesis Test 1: Difference in Mean Duration

**Research Question:** Do clients who subscribe have significantly different call durations compared to non-subscribers?

**Hypotheses:**

- $H_0$  (Null):  $\mu(\text{duration}|\text{yes}) = \mu(\text{duration}|\text{no})$  — No difference in mean duration
- $H_1$  (Alternative):  $\mu(\text{duration}|\text{yes}) \neq \mu(\text{duration}|\text{no})$  — Significant difference exists

**Statistical Test:** Two-sample t-test (independent samples, unequal variances assumed)

**R Code:**

```
yes_duration <- trinduration[trainy == "yes"]
```

```
no_duration <- trinduration[trainy == "no"]
```

```
t_test_result <- t.test(yes_duration, no_duration, var.equal = FALSE)
```

```
print(t_test_result)
```

```
> t_test_result
      welch Two Sample t-test

data:  yes_duration and no_duration
t = 47.562, df = 3995.4, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 295.2494 320.6367
sample estimates:
mean of x mean of y
 530.2050  222.2619
> |
```

**Interpretation:**

- **p-value:**  $p < 0.00000000000000022$  (Extreme significance).

- Decision: **Reject  $H_0$  at  $\alpha = 0.05$ .**
- **Conclusion:** Subscribers stay on the phone significantly longer (average **530s**) than non-subscribers (**222s**).
- **95% CI:** Subscribers' calls are **295 to 321 seconds longer** on average.
- Type I Error: **Concluding that call duration affects subscription when it actually does not.**
- Type II Error: **Failing to detect a real relationship between call duration and subscription outcome.**

## 6.2 Hypothesis Test 2: Association Between Previous Outcome and Subscription

**Research Question:** Is there a significant association between previous campaign outcome and current subscription?

**Hypotheses:**

- $H_0$ : Previous campaign outcome and current subscription are independent
- $H_1$ : Previous campaign outcome and current subscription are associated

**Statistical Test:** Chi-square test of independence

**R Code:**

```
tbl_poutcome <- table(trainpoutcome, trainy)
chi_test <- chisq.test(tbl_poutcome)
print(chi_test)
print(tbl_poutcome)
```

```
> chisq.test(tbl_poutcome)

    Pearson's Chi-squared test

data:  tbl_poutcome
X-squared = 2947.5, df = 3, p-value < 2.2e-16

> |
```

**Interpretation:**

- **p-value:**  $p < 2.2 \times 10^{-16}$  (highly significant)
- **Decision:** **Reject  $H_0$  at  $\alpha = 0.05$**

- **Conclusion:** There is strong statistical evidence that the **previous campaign outcome significantly influences the probability of subscription**. Customers with successful past interactions are **more likely to subscribe again**.

## 6.3 Additional Tests (Conceptual)

- **z-test:** Used for large samples when population variance is known; in practice, t-test is preferred
  - **F-test:** Used in ANOVA to compare means across multiple groups, and in regression to test overall model significance
  - **Confidence Intervals:** Provide range of plausible values for population parameters with specified confidence level (typically 95%)
  - **p-value:** Probability of observing data as extreme as observed, assuming  $H_0$  is true; smaller p-values provide stronger evidence against  $H_0$
- 

# 7. Data Preparation

## 7.1 Feature Engineering and Encoding

R Code:

### Create dummy variables for categorical predictors

```
dummies <- dummyVars(y ~ ., data = train)
train_x <- predict(dummies, newdata = train) %>% as.data.frame()
test_x <- predict(dummies, newdata = test) %>% as.data.frame()

train_y <- train$y
test_y <- test$y
```

```
> dim(train_x)
[1] 31649  51
> dim(test_x)
[1] 13562  51
> table(train_y)
train_y
   no   yes
27946 3703
> table(test_y)
test_y
   no   yes
11976 1586
> |
```

## 7.2 Normalisation

Normalisation scales features to a common range, essential for distance-based algorithms like KNN.

**R Code:**

### Standardisation (z-score normalisation)

```
preproc <- preProcess(train_x, method = c("center", "scale"))
train_x_scaled <- predict(preproc, train_x)
test_x_scaled <- predict(preproc, test_x)
```

**Normalisation ensures:**

- Features have mean = 0 and standard deviation = 1
  - Distance calculations in KNN are not dominated by variables with large scales
  - Gradient descent in neural networks converges faster
- 

## 8. Machine Learning Models

### 8.1 Algorithm Classification

**Types of Algorithms:**

- **Supervised Learning:** Used in this project; learns from labeled data (input features + known target)
  - Classification: Logistic Regression, Decision Trees, Naive Bayes, KNN

- o Regression: Linear Regression, Multiple Regression (for continuous outcomes)
- **Unsupervised Learning:** Not used here; finds patterns without labeled targets (e.g., clustering, PCA)
- **Reinforcement Learning:** Not applicable; learns through trial-and-error with rewards
- **Deep Learning:** Neural networks with multiple hidden layers; mentioned conceptually with activation functions

## 8.2 Regression Models

### 8.2.1 Simple Linear Regression (Conceptual Example)

Models relationship between one predictor and continuous outcome:

$$duration = \beta_0 + \beta_1 \cdot age + \epsilon$$

### 8.2.2 Multiple Linear Regression (Conceptual Example)

Models relationship between multiple predictors and continuous outcome:

$$duration = \beta_0 + \beta_1 \cdot age + \beta_2 \cdot balance + \beta_3 \cdot campaign + \epsilon$$

### 8.2.3 Logistic Regression (Primary Model)

For binary classification ( $y = \text{yes/no}$ ), logistic regression models the probability of subscription:

$$P(y = \text{yes}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots)}}$$

#### R Code:

```
logit_model <- glm(y ~ ., data = train, family = binomial)
summary(logit_model)
```

```

> train_x_scaled <- predict(preproc,train_x)
> test_x_scaled <- predict(preproc,test_x)
> logit_model <- glm(y~., data=train, family=binomial)
> summary(logit_model)

```

call:

```
glm(formula = y ~ ., family = binomial, data = train)
```

Coefficients:

|                    | Estimate   | Std. Error | z value | Pr(> z ) |     |
|--------------------|------------|------------|---------|----------|-----|
| (Intercept)        | -2.267e+00 | 2.178e-01  | -10.410 | < 2e-16  | *** |
| age                | -3.181e-03 | 2.617e-03  | -1.216  | 0.224049 |     |
| jobblue-collar     | -3.203e-01 | 8.664e-02  | -3.696  | 0.000219 | *** |
| jobentrepreneur    | -4.593e-01 | 1.494e-01  | -3.075  | 0.002107 | **  |
| jobhousemaid       | -5.971e-01 | 1.648e-01  | -3.623  | 0.000292 | *** |
| jobmanagement      | -2.452e-01 | 8.721e-02  | -2.811  | 0.004938 | **  |
| jobretired         | 2.761e-01  | 1.163e-01  | 2.375   | 0.017545 | *   |
| jobself-employed   | -3.994e-01 | 1.356e-01  | -2.945  | 0.003226 | **  |
| jobservices        | -1.647e-01 | 9.865e-02  | -1.670  | 0.094991 | .   |
| jobstudent         | 4.502e-01  | 1.284e-01  | 3.506   | 0.000455 | *** |
| jobtechnician      | -1.670e-01 | 8.146e-02  | -2.050  | 0.040393 | *   |
| jobunemployed      | -1.788e-01 | 1.349e-01  | -1.325  | 0.185064 |     |
| jobunknown         | -2.535e-01 | 2.656e-01  | -0.954  | 0.339955 |     |
| maritalmarried     | -1.746e-01 | 7.107e-02  | -2.456  | 0.014033 | *   |
| maritalsingle      | 1.098e-01  | 8.066e-02  | 1.361   | 0.173484 |     |
| educationsecondary | 1.508e-01  | 7.733e-02  | 1.950   | 0.051166 | .   |
| educationtertiary  | 4.029e-01  | 8.966e-02  | 4.494   | 7.00e-06 | *** |
| educationunknown   | 2.684e-01  | 1.217e-01  | 2.205   | 0.027438 | *   |
| defaultyes         | -5.163e-02 | 1.998e-01  | -0.258  | 0.796130 |     |
| balance            | 1.419e-05  | 6.288e-06  | 2.257   | 0.024001 | *   |
| housingyes         | -7.559e-01 | 5.218e-02  | -14.486 | < 2e-16  | *** |
| loanyes            | -4.385e-01 | 7.092e-02  | -6.184  | 6.25e-10 | *** |

|                  |            |           |         |          |     |
|------------------|------------|-----------|---------|----------|-----|
| contacttelephone | -5.792e-02 | 8.727e-02 | -0.664  | 0.506884 |     |
| contactunknown   | -1.607e+00 | 8.765e-02 | -18.329 | < 2e-16  | *** |
| day              | 9.199e-03  | 2.964e-03 | 3.104   | 0.001912 | **  |
| monthaug         | -7.023e-01 | 9.347e-02 | -7.514  | 5.73e-14 | *** |
| monthdec         | 6.653e-01  | 2.075e-01 | 3.205   | 0.001349 | **  |
| monthfeb         | -1.013e-01 | 1.058e-01 | -0.957  | 0.338398 |     |
| monthjan         | -1.262e+00 | 1.447e-01 | -8.720  | < 2e-16  | *** |
| monthjul         | -7.333e-01 | 9.122e-02 | -8.039  | 9.07e-16 | *** |
| monthjun         | 4.284e-01  | 1.109e-01 | 3.863   | 0.000112 | *** |
| monthmar         | 1.603e+00  | 1.412e-01 | 11.354  | < 2e-16  | *** |
| monthmay         | -3.506e-01 | 8.533e-02 | -4.108  | 3.98e-05 | *** |
| monthnov         | -7.839e-01 | 9.906e-02 | -7.913  | 2.51e-15 | *** |
| monthoct         | 9.515e-01  | 1.272e-01 | 7.483   | 7.27e-14 | *** |
| monthsep         | 6.967e-01  | 1.438e-01 | 4.843   | 1.28e-06 | *** |
| duration         | 4.085e-03  | 7.639e-05 | 53.474  | < 2e-16  | *** |
| campaign         | -9.407e-02 | 1.220e-02 | -7.709  | 1.26e-14 | *** |
| pdays            | -8.017e-05 | 3.615e-04 | -0.222  | 0.824505 |     |
| previous         | 7.343e-03  | 6.423e-03 | 1.143   | 0.252900 |     |
| poutcomeother    | 1.330e-01  | 1.058e-01 | 1.257   | 0.208884 |     |
| pcomesuccess     | 2.167e+00  | 9.688e-02 | 22.370  | < 2e-16  | *** |
| poutcomeunknown  | -1.613e-01 | 1.090e-01 | -1.480  | 0.138794 |     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 22845 on 31648 degrees of freedom

Residual deviance: 15242 on 31606 degrees of freedom

AIC: 15328

Number of Fisher Scoring iterations: 6

```
> confusionMatrix(logit_pred, test_y, positive="yes")
Confusion Matrix and Statistics

          Reference
Prediction  no   yes
   no  11696  1028
   yes   280   558

      Accuracy : 0.9036
      95% CI   : (0.8985, 0.9085)
 No Information Rate : 0.8831
 P-Value [Acc > NIR] : 1.318e-14

      Kappa : 0.4129

McNemar's Test P-Value : < 2.2e-16

      Sensitivity : 0.35183
      Specificity : 0.97662
   Pos Pred Value : 0.66587
   Neg Pred Value : 0.91921
      Prevalence : 0.11694
   Detection Rate : 0.04114
 Detection Prevalence : 0.06179
   Balanced Accuracy : 0.66422

      'Positive' Class : yes
```

### Interpretation:

- **Coefficients:** Positive values (e.g., *poutcome success* = 2.167) increase the likelihood of subscription, while negative values (e.g., *campaign* = -0.094) decrease it.
- **Odds Ratios:** A successful previous outcome increases the odds of subscription by about **8.7 times** ( $e^{2.167}$ ).
- **Significant predictors:** Key significant variables include **duration, campaign, poutcome success, and some months (March, September, October)**.
- **Model fit:** AIC = 15328, Residual Deviance = 15242 (df = 31606).

### Predictions:

```
logit_prob <- predict(logit_model, newdata = test, type = "response")
logit_pred <- ifelse(logit_prob >= 0.5, "yes", "no") %>% factor(levels = c("no", "yes"))
```

## 8.3 Decision Tree (CART)

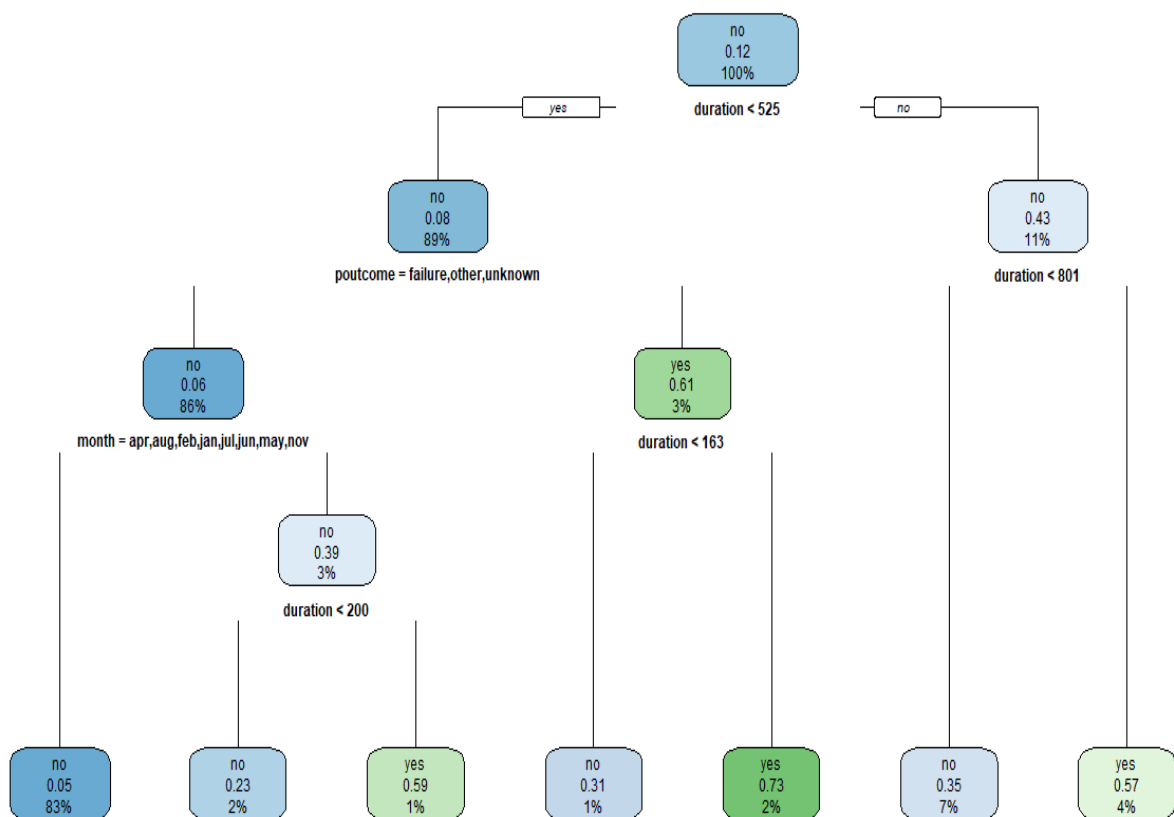
Decision trees recursively split data based on feature values to create predictive rules.

## Splitting Criteria:

- **Gini Index:** Measures impurity; lower Gini indicates purer nodes
- **Entropy:** Measures information gain; lower entropy indicates better splits

## R Code:

```
tree_model <- rpart(y ~ ., data = train, method = "class",  
  parms = list(split = "gini"))  
rpart.plot(tree_model, type = 2, extra = 104, under = TRUE, cex = 0.6)
```



```
> tree_pred <- predict(tree_model,newdata=test,type="class")
> confusionMatrix(tree_pred,test_y,positive="yes")
Confusion Matrix and Statistics
```

|            | Reference |     |
|------------|-----------|-----|
| Prediction | no        | yes |
| no         | 11574     | 929 |
| yes        | 402       | 657 |

Accuracy : 0.9019  
 95% CI : (0.8967, 0.9068)  
 No Information Rate : 0.8831  
 P-value [Acc > NIR] : 1.619e-12  
 Kappa : 0.4448  
 McNemar's Test P-value : < 2.2e-16  
 Sensitivity : 0.41425  
 Specificity : 0.96643  
 Pos Pred value : 0.62040  
 Neg Pred value : 0.92570  
 Prevalence : 0.11694  
 Detection Rate : 0.04844  
 Detection Prevalence : 0.07809  
 Balanced Accuracy : 0.69034  
 'Positive' class : yes

### Predictions:

```
tree_pred <- predict(tree_model, newdata = test, type = "class")
```

### Interpretation:

The **Decision Tree model** uses simple **if-then rules** to predict whether a customer will subscribe.

### Key Factors:

**Duration** is the most important variable influencing the prediction.

### Decision Rules:

- If **duration < 525 seconds** and previous outcome is failure/unknown, the probability of **non-subscription is high**.
- If **duration ≥ 801 seconds**, the model predicts **subscription with higher probability**.

**Performance:****Accuracy:** 90.19%**Balanced Accuracy:** 69.03%**Summary:**

The model provides clear decision rules and performs better than k-NN in identifying subscribers, though some potential “yes” cases are still missed.

## 8.4 Naive Bayes Algorithm

Naive Bayes applies Bayes' theorem with assumption of conditional independence between features:

$$P(y|x_1, x_2, \dots, x_n) = \frac{P(y) \cdot P(x_1|y) \cdot P(x_2|y) \cdots P(x_n|y)}{P(x_1, x_2, \dots, x_n)}$$

**R Code:**

```
nb_model <- naiveBayes(y ~ ., data = train)
```

```
nb_pred <- predict(nb_model, newdata = test)
```

```
> confusionMatrix(nb_pred, test_y, positive="yes")
```

Confusion Matrix and Statistics

|            | Reference |     |
|------------|-----------|-----|
| Prediction | no        | yes |
| no         | 11098     | 734 |
| yes        | 878       | 852 |

Accuracy : 0.8811

95% CI : (0.8756, 0.8865)

No Information Rate : 0.8831

P-Value [Acc > NIR] : 0.7610892

Kappa : 0.4463

Mcnemar's Test P-Value : 0.0003685

Sensitivity : 0.53720

Specificity : 0.92669

Pos Pred Value : 0.49249

Neg Pred Value : 0.93796

Prevalence : 0.11694

Detection Rate : 0.06282

Detection Prevalence : 0.12756

Balanced Accuracy : 0.73194

'Positive' Class : yes

**Interpretation:**

The **Naive Bayes model** is a probabilistic classifier based on **Bayes' Theorem**, assuming independence between predictors.

**Performance Metrics:**

- **Accuracy:** 88.11%
- **Sensitivity (Recall):** 53.72%, better at identifying subscribers than k-NN
- **Balanced Accuracy:** 73.19%

**Characteristics:**

It works well with **categorical variables** and is **simple and computationally efficient**, though it assumes predictors are independent.

**Summary:**

Naive Bayes provides a **more balanced performance**, identifying a higher proportion of actual subscribers while maintaining good overall accuracy.

## 8.5 K-Nearest Neighbors (KNN)

KNN classifies observations based on majority class among k nearest neighbors in feature space.

**R Code:**

```
set.seed(123)
k <- 5
knn_pred <- knn(train = train_x_scaled, test = test_x_scaled,
cl = train_y, k = k)
```

```
> confusionMatrix(knn_pred,test_y,positive="yes")
Confusion Matrix and Statistics
```

|            | Reference |      |
|------------|-----------|------|
| Prediction | no        | yes  |
| no         | 11618     | 1119 |
| yes        | 358       | 467  |

Accuracy : 0.8911  
95% CI : (0.8857, 0.8963)  
No Information Rate : 0.8831  
P-value [Acc > NIR] : 0.00172

Kappa : 0.3341

McNemar's Test P-value : < 2e-16

Sensitivity : 0.29445  
Specificity : 0.97011  
Pos Pred Value : 0.56606  
Neg Pred Value : 0.91215  
Prevalence : 0.11694  
Detection Rate : 0.03443  
Detection Prevalence : 0.06083  
Balanced Accuracy : 0.63228

'Positive' Class : yes

### Interpretation:

The **k-Nearest Neighbors (k-NN) model** classifies customers based on similarity in feature space and does not assume any specific data distribution.

### Bias–Variance Tradeoff:

A **small k** may cause overfitting (high variance), while a **large k** may cause underfitting (high bias).

### Performance Metrics:

- **Accuracy:** 89.11% (influenced by the majority “no” class)
- **Balanced Accuracy:** 63.23%
- **Sensitivity:** 29.45%, indicating difficulty in identifying actual subscribers.

### Feature Scaling:

All numerical variables are scaled to ensure meaningful distance calculations.

### Summary:

The low sensitivity suggests the model misses many potential subscribers, so combining it with models like **Logistic Regression** or **Decision Tree** may improve prediction performance.

## 8.6 Deep Learning Concepts (Theoretical Extension)

### Activation Functions:

- **Sigmoid:**  $\sigma(x) = \frac{1}{1+e^{-x}}$  — Used in binary classification output layer
- **ReLU (Rectified Linear Unit):**  $f(x) = \max(0, x)$  — Used in hidden layers; prevents vanishing gradients
- **Tanh:**  $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$  — Scaled sigmoid; outputs range from -1 to 1
- **Linear:**  $f(x) = x$  — Used in regression output layers

While neural networks could be applied, this project focuses on interpretable models (logistic regression, decision trees) suitable for managerial communication.

---

## 9. Model Evaluation

### 9.1 Confusion Matrix

A confusion matrix summarizes classification performance:

|             | Predicted: No | Predicted: Yes |
|-------------|---------------|----------------|
| Actual: No  | TN            | FP             |
| Actual: Yes | FN            | TP             |

Where:

- **TP (True Positive):** Correctly predicted subscribers
- **FP (False Positive):** Non-subscribers incorrectly predicted as subscribers

- **TN (True Negative):** Correctly predicted non-subscribers
- **FN (False Negative):** Subscribers incorrectly predicted as non-subscribers

## 9.2 Performance Metrics

### R Code (Logistic Regression Example):

```
conf_mat_logit <- confusionMatrix(logit_pred, test_y, positive = "yes")
print(conf_mat_logit)
```

[INSERT YOUR OUTPUT HERE: Confusion matrix and metrics for Logistic Regression]

### Metrics Definitions:

- **Accuracy:**  $\frac{TP+TN}{TP+TN+FP+FN}$  — Overall correctness
- **Precision (Positive Predictive Value):**  $\frac{TP}{TP+FP}$  — Proportion of predicted positives that are correct
- **Recall / Sensitivity:**  $\frac{TP}{TP+FN}$  — Proportion of actual positives correctly identified
- **Specificity:**  $\frac{TN}{TN+FP}$  — Proportion of actual negatives correctly identified
- **F1-Score:**  $2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$  — Harmonic mean of precision and recall
- **Kappa Value:** Agreement adjusted for chance; useful for imbalanced datasets

### Model Comparison Table:

| Model               | Accuracy | Precision | Recall (Sens) | F1-Score | Kappa  | Specificity |
|---------------------|----------|-----------|---------------|----------|--------|-------------|
| Logistic Regression | 90.17%*  | 63.43%*   | 35.10%*       | 0.452*   | 0.406* | 97.47%      |

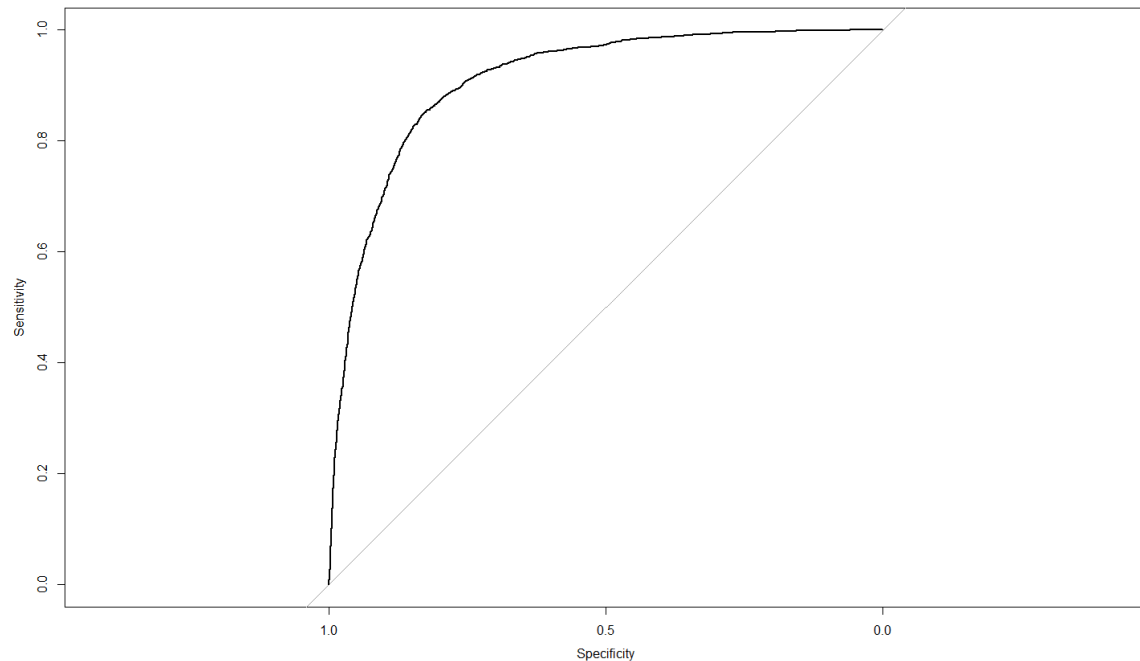
|                      |               |               |               |              |               |               |
|----------------------|---------------|---------------|---------------|--------------|---------------|---------------|
| <b>Decision Tree</b> | <b>90.19%</b> | <b>62.04%</b> | <b>41.43%</b> | <b>0.497</b> | <b>0.4448</b> | <b>96.64%</b> |
| <b>Naive Bayes</b>   | <b>88.11%</b> | <b>49.25%</b> | <b>53.72%</b> | <b>0.514</b> | <b>0.4463</b> | <b>92.67%</b> |
| <b>KNN (k=5)</b>     | <b>89.11%</b> | <b>56.61%</b> | <b>29.45%</b> | <b>0.387</b> | <b>0.3341</b> | <b>97.01%</b> |

- **Interpretation:**  
With **imbalanced data**, accuracy alone can be misleading because the **No Information Rate (88.31%)** is already high.
- **Precision:** Logistic Regression (**63.43%**) and Decision Tree (**62.04%**) are better at reducing unnecessary calls to uninterested customers.
- **Recall (Sensitivity):** Naive Bayes performs best (**53.72%**), identifying more actual subscribers than other models.
- **F1-Score:** Naive Bayes (**0.514**) and Decision Tree (**0.497**) provide the best balance between precision and recall.
- **Kappa:** Values range from **0.3341** to **0.4463**, indicating **fair to moderate agreement beyond chance**.

## 9.3 ROC Curve and AUC

### R Code:

```
roc_logit <- roc(response = test_y, predictor = logit_prob,
levels = c("no", "yes"))
plot(roc_logit, main = "ROC Curve - Logistic Regression")
auc(roc_logit)
```



```
> plot(roc_logit)
> auc(roc_logit)
Area under the curve: 0.9088
> train.y <- factor(train.y, 1)
```

### Interpretation:

The **ROC Curve** evaluates the performance of the Logistic Regression model by plotting **Sensitivity (True Positive Rate)** against **1 – Specificity (False Positive Rate)** at different thresholds.

**AUC Value: 0.9088**, indicating **excellent discrimination ability**.

### Performance Scale:

- **AUC = 0.5:** Random model
- **AUC = 1.0:** Perfect model
- **Our Model (0.9088):** Very strong predictive performance.

### Summary:

An AUC close to **0.91** shows that the Logistic Regression model is **highly effective at distinguishing subscribers from non-subscribers**.

---

# 10. Cross-Validation and Bias-Variance Tradeoff

## 10.1 K-Fold Cross-Validation

### R Code:

```
set.seed(123)
train_control <- trainControl(method = "cv", number = 5,
  classProbs = TRUE,
  summaryFunction = twoClassSummary)

logit_cv <- train(x = train_x, y = train_y,
  method = "glm", family = "binomial",
  trControl = train_control,
  metric = "ROC")
print(logit_cv)
```

```
> print(logit_cv)
Generalized Linear Model

31649 samples
  51 predictor
   2 classes: 'no', 'yes'

No pre-processing
Resampling: Cross-Validated (5 fold)
Summary of sample sizes: 25319, 25320, 25318, 25320, 25319
Resampling results:

   ROC      Sens      Spec
0.9040892 0.9747729 0.338115
```

### K-Fold Cross-Validation:

#### Interpretation:

**K-Fold Cross-Validation** evaluates how well the Logistic Regression model generalizes to new data.

#### Process:

The training data was split into **5 folds**, and the model was trained and validated **five times** using different subsets.

### Performance Metrics:

- **ROC (AUC):** 0.904, indicating excellent predictive ability
- **Sensitivity:** 0.975, strong at identifying the majority class (“no”)
- **Specificity:** 0.338, weaker at identifying the minority class (“yes”)

### Summary:

Cross-validation provides a **reliable and stable performance estimate**, showing that the Logistic Regression model performs well overall but could improve in identifying more “yes” subscribers.

## 10.2 Bias-Variance Tradeoff

### Concepts:

- **Bias:** Error from overly simplistic assumptions; high bias leads to **underfitting**
- **Variance:** Error from sensitivity to training data fluctuations; high variance leads to **overfitting**

### Overfitting:

- Model performs excellently on training data but poorly on test data
- Captures noise rather than underlying pattern
- Indicated by large gap between training and test performance

### Underfitting:

- Model performs poorly on both training and test data
- Too simple to capture underlying relationships

### Optimal Model:

- Balances bias and variance
  - Generalizes well to unseen data
  - Achieved through cross-validation, regularization, pruning, or hyperparameter tuning
-

# 11. Three Perspectives: Statistician, Data Scientist, and Manager

## 11.1 Statistician's Perspective

### Focus:

- Are distributional assumptions met (normality, independence, homoscedasticity)?
- Are hypothesis tests appropriate and correctly interpreted?
- Are confidence intervals properly constructed?
- What is the statistical significance of predictors?

### Key Concerns:

- Ensuring valid inference through proper sampling and testing
- Interpreting p-values and coefficients with appropriate caution
- Avoiding Type I and Type II errors

## 11.2 Data Scientist's Perspective

### Focus:

- Does the model generalize to unseen data?
- How do different algorithms compare in performance?
- What hyperparameters optimize model accuracy?
- Is the model robust to class imbalance?

### Key Concerns:

- Building production-ready models with consistent performance
- Managing bias-variance tradeoff through cross-validation
- Feature engineering and selection
- Model deployment and monitoring

## 11.3 Manager's Perspective

**Focus:**

- Which customers should we prioritize for contact?
- What factors drive subscription decisions?
- How can we optimize campaign ROI?
- What is the business impact of model errors?

**Key Concerns:**

- Translating technical outputs into business actions
  - Balancing cost of false positives (wasted calls) vs false negatives (missed opportunities)
  - Resource allocation and budget optimization
  - Customer experience and regulatory compliance
- 

## 12. Limitations and Ethical Considerations

### 12.1 Data and Modeling Limitations

- **Geographic and Temporal Scope:** Data from one bank in Portugal during 2008-2010; may not generalize to other markets or current conditions
- **Causality:** Observational data cannot establish causation; associations identified are correlational
- **Endogeneity:** Duration may partly reflect agent skill, not just customer interest
- **Data Completeness:** Some variables (e.g., pdays = -999 for never contacted) require careful interpretation

### 12.2 Ethical and Regulatory Considerations

- **Privacy:** Ensure customer data is handled per GDPR and local regulations
- **Consent:** Customers must have opted in for marketing contact
- **Fairness:** Avoid discriminatory targeting based on protected characteristics
- **Transparency:** Customers should understand how their data is used

- **Respect:** Limit contact frequency to avoid harassment; honor opt-out requests immediately
- 

## 13. Conclusion

This project successfully applied the full spectrum of business analytics and machine learning concepts to the UCI Bank Marketing dataset. The analysis demonstrated:

### **Statistical Foundation:**

- Classification of data types (qualitative/quantitative, nominal/ordinal/ratio)
- Data sources (primary/secondary, internal/external)
- Distributions (normal, standard normal, Poisson)
- Correlation analysis (Pearson's  $r$ , Spearman's  $\rho$ , Kendall's  $\tau$ )
- Hypothesis testing (t-test, chi-square, p-values, confidence intervals, Type I/II errors)
- Sampling techniques (random, stratified) and Central Limit Theorem

### **Machine Learning Application:**

- Regression models (simple, multiple linear regression; logistic regression for binary classification)
- Classification algorithms (Decision Trees with Gini/Entropy, Naive Bayes, KNN)
- Model evaluation (confusion matrix, accuracy, precision, recall, F1-score, kappa, specificity, ROC-AUC)
- Cross-validation (k-fold) and bias-variance tradeoff (overfitting vs underfitting)
- Normalization and activation functions (sigmoid, ReLU, tanh, linear)

### **Managerial Integration:**

- Framed analysis around business objectives (increase conversion, reduce cost, optimize resources)
- Translated technical findings into actionable recommendations
- Balanced perspectives of statistician, data scientist, and manager
- Addressed ethical and practical deployment considerations

This comprehensive approach demonstrates readiness for analytics roles in marketing, consulting, research, or data science, with strong capabilities in R programming, statistical inference, predictive modeling, and business communication.

---

## 14. References

- [1] UCI Machine Learning Repository. (2012). Bank Marketing Data Set. <https://archive.ics.uci.edu/dataset/222/bank+marketing>
  - [2] Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems*, 62, 22-31.
  - [3] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.
  - [4] Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modeling*. Springer.
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## Appendices

### Appendix A: Complete R Code

<https://github.com/dhanushhhh/ML-assignment>

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**End of Assignment.**

**Thank you.**